

## Vertical dilations



1 Consider the expression  $\cos \theta$ .

a Complete the table of values for different values of  $\theta$ .

| $\theta$      | 0 | 60 | 90 | 120 | 180 | 240 | 270 | 300 | 360 |
|---------------|---|----|----|-----|-----|-----|-----|-----|-----|
| $\cos \theta$ |   |    |    |     |     |     |     |     |     |

b Sketch a graph of the function  $y = \cos \theta$ .

c State the maximum value of  $\cos \theta$ .

d State the minimum value of  $\cos \theta$ .

e State the range of values of  $4 \cos \theta$ .

2 Consider the functions  $y = \sin x$  and  $y = \cos x$ .

a State the domain of both functions.

b State the range of both functions.

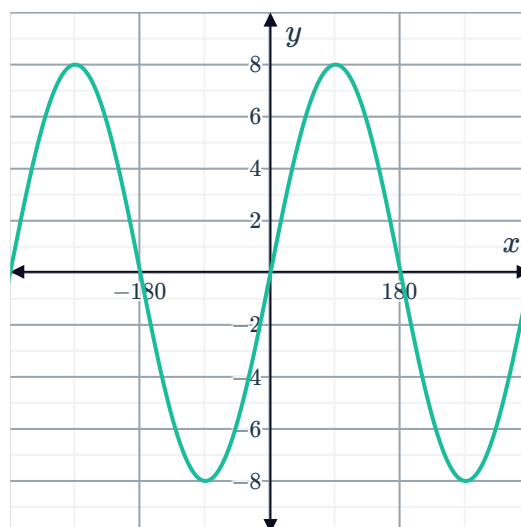
3 Consider the function  $y = -2 \cos x$ .

a State the domain of the function.

b State the range of the function.

4 Consider the given graph of a function of the form  $f(x) = A \sin x$ :

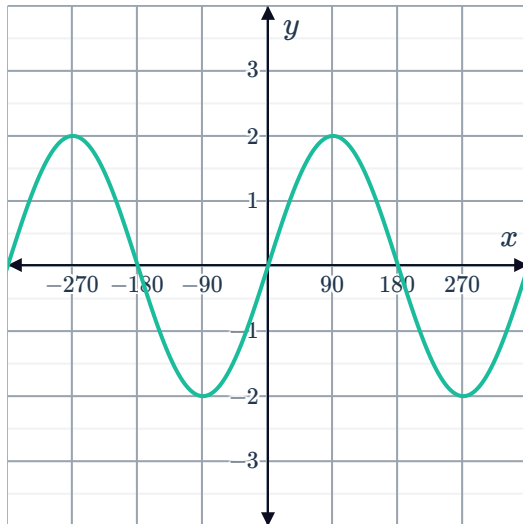
State the amplitude of the function.



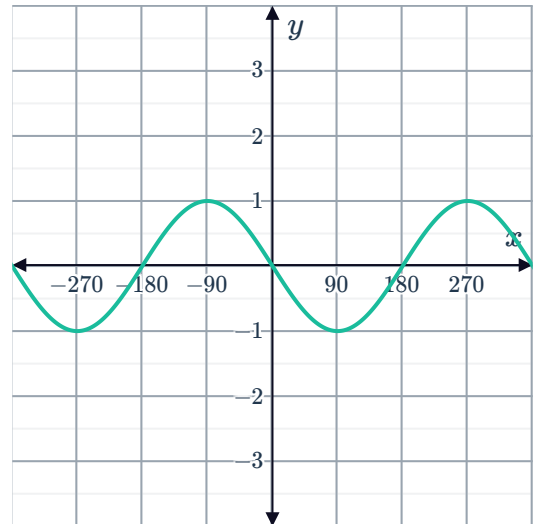
- 5 State the equation of each of the functions plotted below given that they are of the form  $y = a \sin x$  or  $y = a \cos x$ :



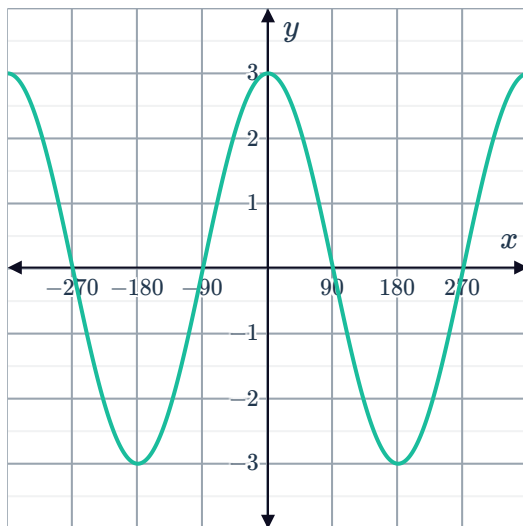
a



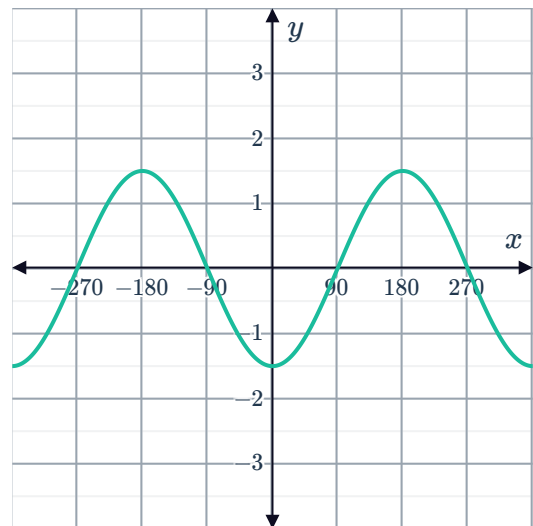
b



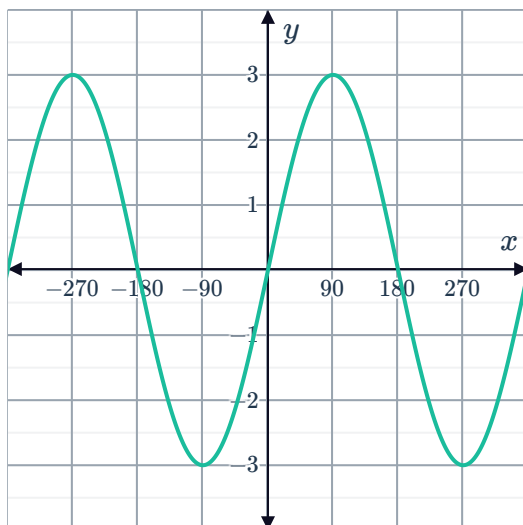
c



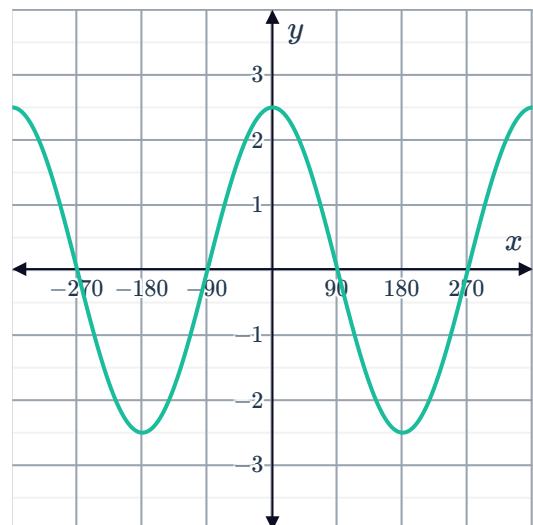
d



e



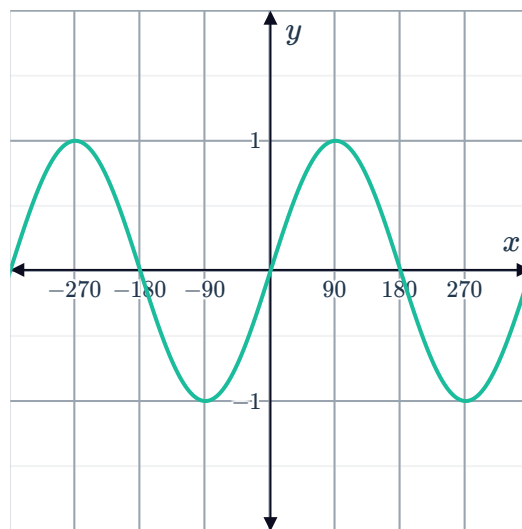
f



- 6 Consider the graph of the function  $y = \sin x$  for  $0 \leq x < 360$ :



At which value of  $x$  in the given domain would  $y = -\sin x$  have a maximum value?

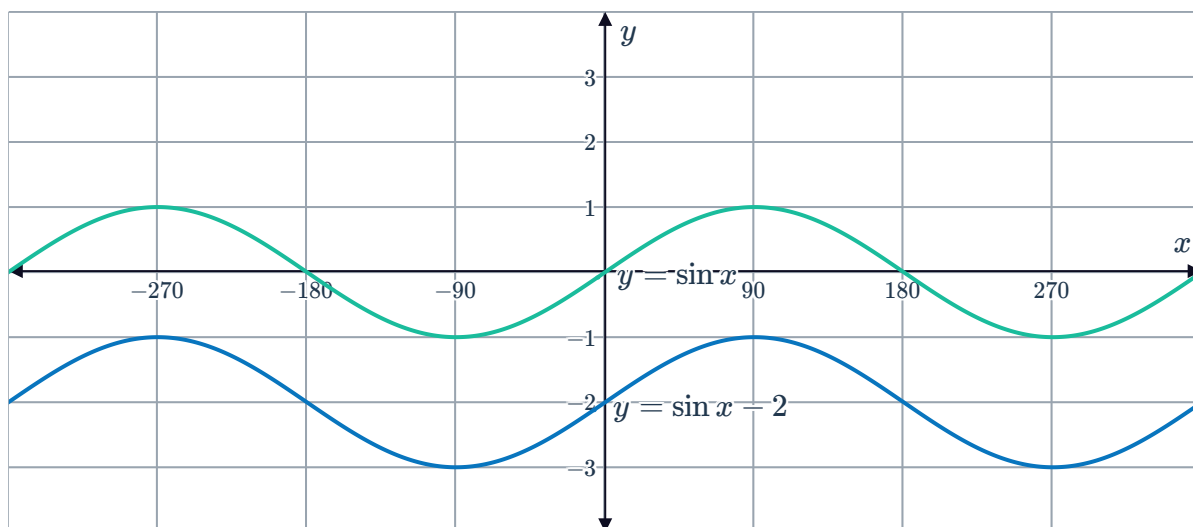


- 7 Consider the function  $f(x) = -8 \sin x$ , where  $0 \leq x \leq 180$ .
- State the amplitude of the function.
  - Find the value of  $f(180)$ .
  - Find the minimum value of the function.
- 8 The function  $y = k \sin x$  has a maximum value of 5. Find the value of  $k$ , where  $k > 0$ .
- 9 A sine function has the form  $y = c \sin x$ , a range of  $[-2, 2]$  and a minimum at  $90$ . Find an expression for  $y$ .
- 10 For each of following functions:
- State the amplitude.
  - Sketch a graph of the function.
- $y = 3 \sin x$ .
  - $y = 4 \cos x$
  - $y = \frac{4}{3} \cos x$
  - $y = -5 \cos x$
  - $y = \frac{5}{4} \sin x$
  - $y = -4 \sin x$
- 11 Consider the function  $y = -3 \cos x$ .
- State the maximum value of the function.
  - State the minimum value of the function.
  - State the amplitude of the function.
  - Describe the two transformations required to obtain the graph of  $y = -3 \cos x$  from the graph of  $y = \cos x$ .



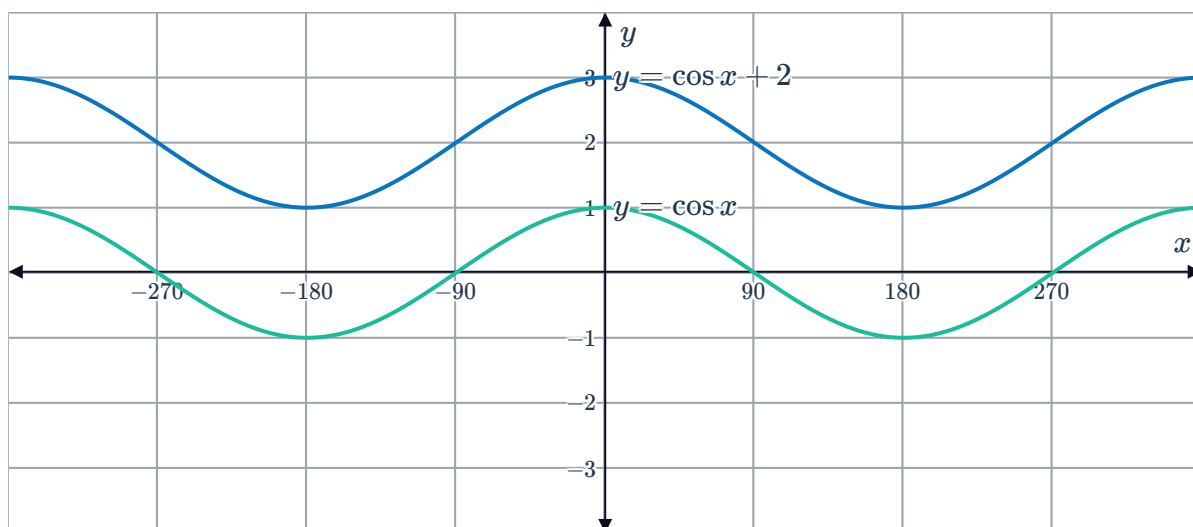
## Vertical translations

12 Consider the two graphs  $y = \sin x$  and  $y = \sin x - 2$  below:



Describe the transformation required to obtain the graph of  $y = \sin x - 2$  from  $y = \sin x$ .

13 Consider the two graphs  $y = \cos x$  and  $y = \cos x + 2$  below:

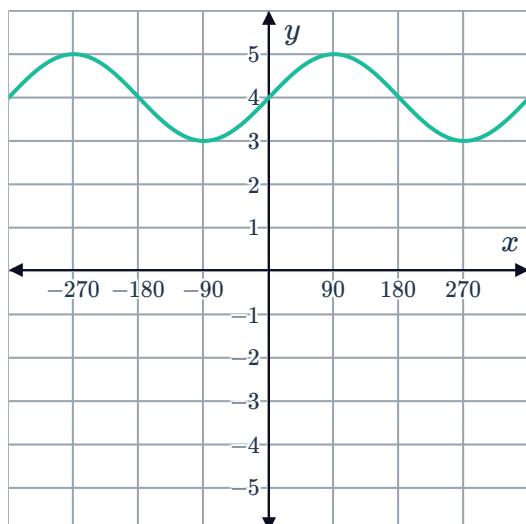


Describe the transformation required to obtain the graph of  $y = \cos x + 2$  from  $y = \cos x$ .

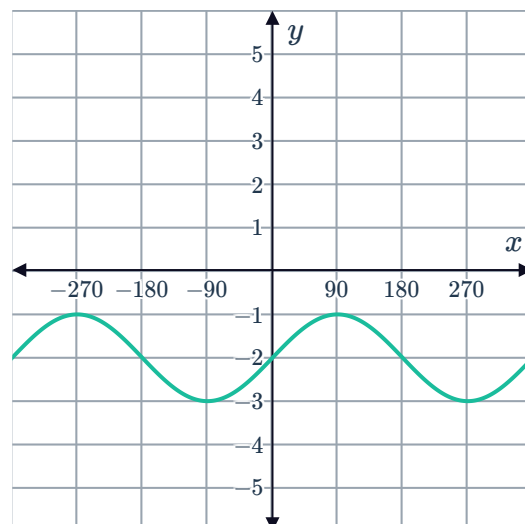
14 Describe the transformation required to obtain the graph of  $y = \sin x + 4$  from  $y = \sin x$ .

- 15 State the equation of each of the functions plotted below given that they are of the form  $y = \sin x + k$ :

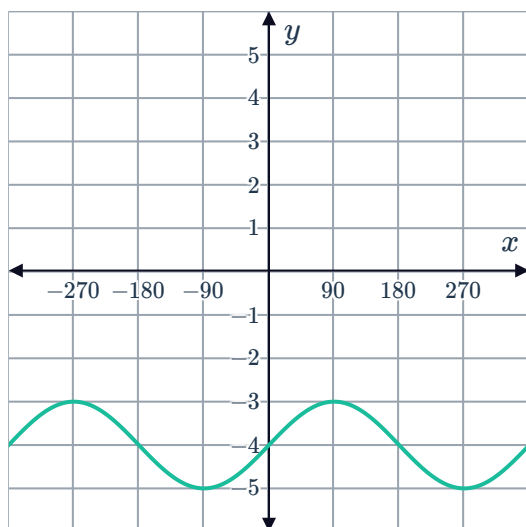
a



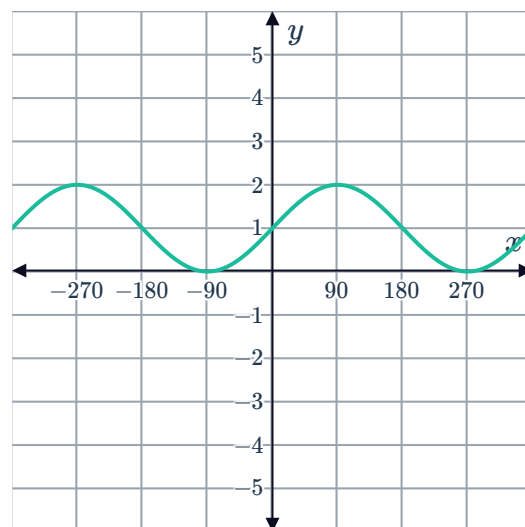
b



c

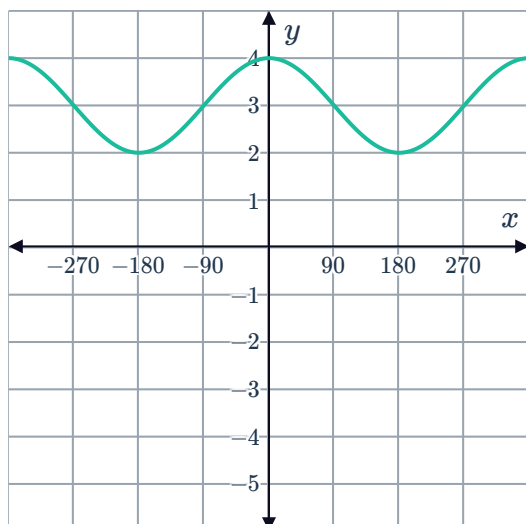


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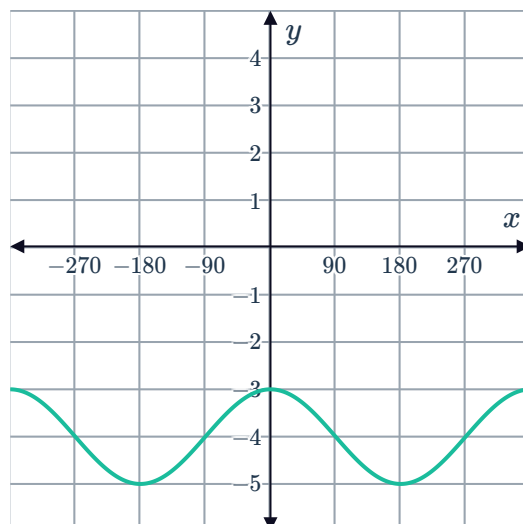


- 16 State the equation of each of the functions graphed below given that they are of the form  $y = \cos x + k$ :

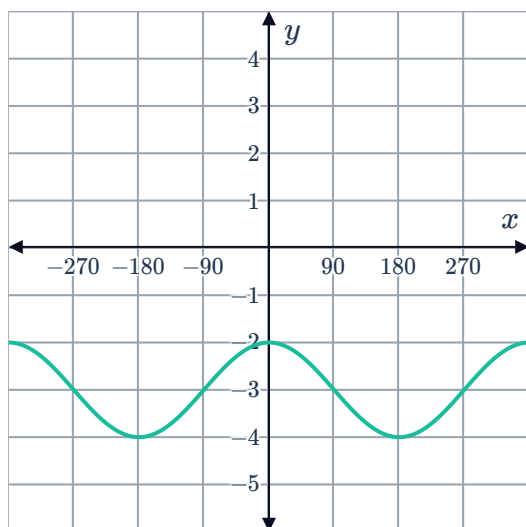
a



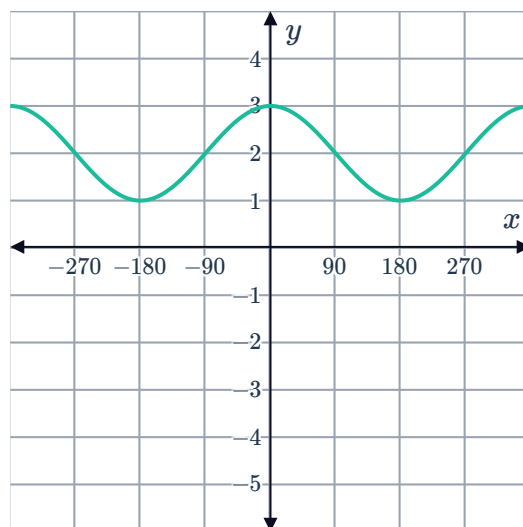
b



c



d



- 17 Describe the two transformations required to turn the graph of  $y = \cos x$  into the graph of  $y = -\cos x + 3$ .

- 18 State the equation of the line that the function  $y = \sin x + 2$  oscillates about.

- 19 Consider the function  $y = \cos x + 4$ .

- Describe the transformation required to obtain the graph of  $y = \cos x + 4$  from  $y = \cos x$ .
- State the period of the function.
- Find the maximum value of the function.
- Find the minimum value of the function.

**20** For each of the functions below:



- i State the period of the function.
- ii State the amplitude of the function.
- iii Find the maximum value of the function.
- iv Find the minimum value of the function.
- v Sketch a graph of the function for  $-360 \leq x \leq 360$ .

**a**  $y = \sin x + 1$       **b**  $y = \cos x - 3$       **c**  $y = \sin x - 2$       **d**  $y = 3 \sin x + 2$   
**e**  $y = 3 \cos x - 3$       **f**  $y = 2 \sin x - 3$

**21** For each of the functions below:

- i Find the value of  $y$  when  $x = 90$ .
- ii State the amplitude of of the function.
- iii Find the maximum value of the function.
- iv Find the minimum value of the function.
- v Describe the transformations required to obtain the graph of the function from  $y = \cos x$ .
- vi Sketch a graph of the function for  $0 \leq x \leq 360$ .

**a**  $y = 2 \cos x + 3$       **b**  $y = 2 \cos x - 3$       **c**  $y = -4 \cos x + 2$       **d**  $y = \frac{1}{3} \sin x + 2$

**22** The function  $y = \cos x + 5$  is translated 4 units up.

- a** State the equation of the new function after the translation.
- b** Find the maximum value of the new function.

**23** For each of the functions below:

- i Find the maximum value of the function.
- ii Find the minimum value of the function.

**a**  $y = 2 - 3 \sin x$       **b**  $y = 4 - 3 \sin x$       **c**  $y = (\sin x)^2 + 1$       **d**  $y = \frac{4}{3 \cos x + 5}$

**24** A sine function,  $y$ , has the form  $y = c \sin x + d$  and a range of  $[0, 4]$ . Find an expression for  $y$ , where  $c > 0$ .

**25** A cosine function,  $y$ , has the form  $y = c \cos x - d$  and a range of  $[-10, 6]$ . Find an expression for  $y$ , where  $c > 0$ .